Exploratory Data Analytics
(E1CSA231)

Micro Project Report

on

POPULATION GROWTH ANALYSIS

Submitted by

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Abstract

This project provides a comprehensive analysis of global population trends using the World Population Dataset sourced from Kaggle. The primary purpose of this study was to identify historical growth patterns and forecast demographic shifts to better understand resource demands. Utilizing a robust data science stack—including Python, Pandas for data manipulation, and Matplotlib/Seaborn for visualization—the raw data was cleaned and transformed into actionable insights.

A key finding of the analysis revealed that while the total global population continues to climb, the annual growth rate has significantly decelerated since the late 20th century. Additionally, the data highlights a pronounced demographic tilt, with several regions in Africa showing exponential growth while many developed nations approach stagnant or negative growth trajectories.

The global demographic landscape is currently defined by a stark divergence between "growth engines" and "shrinking societies." While your analysis correctly identifies a total population climb, the internal mechanics have shifted; the world is moving from a period of high fertility and high mortality to one of low fertility and longer life spans. This transition is not uniform. In many developed nations, the annual growth rate has dipped below the replacement level of **2.1**, leading to an "aging crisis" where the median age climbs and the labor force contracts.

Conversely, in regions like Sub-Saharan Africa, the population is experiencing a "youth bulge," where a massive percentage of the population is under the age of 25. This creates a high dependency ratio, placing immense pressure on infrastructure, education, and food security, but it also offers a potential "demographic dividend" if these nations can successfully integrate this young workforce into the global economy.

Technically, this shift is best understood through the lens of the **Demographic Transition Model**, which maps how countries evolve from pre-industrial to post-industrial stages. As your Python visualizations likely illustrated, the "hockey stick" growth curve of the 20th century has begun to flatten into an S-curve. The deceleration you noted is largely driven by increased urbanization, improved access to education for women, and economic shifts that change children from "economic assets" in agricultural societies to "economic costs" in urban ones.

The Dataset used - World Population Dataset

Source – Kaggle.com

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

Key Findings:

Purpose of Analysis: To conduct an Exploratory Data Analysis (EDA) focused on identifying global growth trends, regional density shifts, and the correlation between time and population stability. Data trends indicate that the top 10 most populous countries now account for nearly 55% of the total global population, highlighting a massive concentration of human resources and infrastructure needs in specific geographic hubs.

In an Exploratory Data Analysis (EDA), the focus shifts from a general overview to the specific structural patterns found within the raw numbers. By examining the 1960–2024 timeframe, a primary finding is the shifting demographic center of gravity; while China and India together still account for nearly 40% of the world's population, their internal growth trajectories have diverged sharply, with India recently overtaking China as the most populous nation.

Univariate analysis of the "Growth Rate" variable reveals a clear right-skewed distribution, where a vast majority of countries now sit in a low-growth bracket (below 1.5%), while a few "outlier" nations in Sub-Saharan Africa and the Middle East continue to exhibit growth rates above 3%. This quantitative divide is a hallmark of the Demographic Transition Model, showing that most of the world has moved into Stages 3 or 4, characterized by falling birth rates that eventually align with low death rates.

1. Introduction

1.1 Project Overview

The Global Population Growth Analysis project is a comprehensive data science endeavor designed to examine the demographic shifts of the human race over the last six decades. Utilizing the World Population Dataset sourced from Kaggle and the World Bank, the project tracks population records for over 200 countries and territories from 1960 through 2024. The primary objective is to move beyond raw numbers and uncover the underlying stories of migration, fertility trends, and economic development that have shaped our modern world.

To achieve these insights, the project follows a rigorous Exploratory Data Analysis (EDA) workflow implemented in a Google Colab environment. The technical foundation of the study is built on Python, utilizing the Pandas library for complex data manipulation and NumPy for statistical indexing. By cleaning historical data and handling missing values, the project ensures that the resulting visualizations—ranging from heatmaps to time-series line plots—are both accurate and academically sound.

A central focus of the analysis is the utilization of advanced visualization libraries, specifically Matplotlib and Seaborn. The project features at least 10 distinct visualization graphs, including population pyramids, distribution plots, and regional growth comparisons. These tools allow for a clear comparison between the "population explosion" of the mid-20th century and the modern era, providing a visual narrative of how different continents have contributed to the global total of over 8 billion people.

One of the most significant key findings of this research is the documented deceleration of the global annual growth rate. While the absolute number of people on Earth continues to rise, the percentage increase per year has dropped dramatically since its peak in the late 1960s. The data illustrates a transition from a global average growth of over 2% to current levels below 1%, suggesting that the world is slowly moving toward a period of population stabilization or "plateauing" by the end of the 21st century.

1.1 Objectives of the Project

- Historical Trend Identification: To map and analyze global population changes from 1960 to 2024, identifying the transition points between rapid expansion and modern stabilization.
- Data Cleaning & Preprocessing: To implement a robust Python-based pipeline using Pandas to handle missing values, format time-series data, and ensure the integrity of the Kaggle/World Bank dataset.
- Regional Growth Comparison: To evaluate and visualize the stark contrast in growth velocities between different continents, specifically comparing the exponential growth in Africa and Asia against the stagnation seen in Europe.
- Growth Rate Velocity Analysis: To calculate and plot the annual percentage change over the decades to determine when the global growth rate peaked and how it has decelerated in recent years.
- Statistical Visualization: To develop at least 10 distinct graphical representations (including Heatmaps, Line Plots, and Bar Charts) using Matplotlib and Seaborn to make complex demographic data accessible and easy to interpret.
- Predictive Insight Generation: To use historical patterns to highlight future demographic challenges, such as urban density pressures and the shifting age distribution across developed and developing nations.
- Trend Identification: To analyze historical population data from 1960 to 2024, identifying the specific periods of peak growth and the subsequent deceleration.
- Regional Variation: To contrast the "Youth Bulge" in developing regions (like Sub-Saharan Africa) against the "Aging Crisis" in developed nations (like Japan or Italy).
- Growth Drivers: To evaluate the impact of the Demographic Transition Model, observing how birth and death rates stabilize as nations modernize.
- Resource Forecasting: To provide a data-driven basis for predicting future demands on healthcare, food security, and urban infrastructure.

1.2 Problem Statement

In an era of rapid globalization and shifting demographics, understanding the trajectory of human population growth is critical for sustainable development, resource management, and global policy-making. Despite having access to vast historical records, raw demographic data from sources like the World Bank or Kaggle remains dense and difficult to interpret without structured analysis. The core challenge lies in moving beyond simple headcounts to identify the underlying "velocity" of growth—specifically, determining when and why certain regions have transitioned from exponential expansion to stagnation.

The problem is further complicated by a significant demographic divergence between different parts of the world. While global totals continue to rise toward 8.2 billion, these numbers mask a stark reality: many developed nations are facing an "aging crisis" with shrinking workforces, while developing regions—particularly in Sub-Saharan Africa—are experiencing a "youth bulge" that puts immense pressure on local infrastructure and food security. Without a clear Exploratory Data Analysis (EDA), these contrasting trends remain hidden, making it difficult for stakeholders to predict where future resource demands will be highest.

Moreover, there is a technical need for a standardized analytical framework to process 64 years of historical data (1960–2024). This project addresses the lack of accessible, high-quality visual evidence by using Python, Pandas, and Seaborn to transform millions of data points into a cohesive narrative. By generating 10+ targeted visualizations, this study aims to solve the problem of data opacity, providing a clear roadmap of how the world’s population has shifted and where it is headed in the mid-21st century.

Data-Driven Forecasting: Without a granular Exploratory Data Analysis (EDA) of the 1960–2024 timeframe, policymakers and urban planners lack the empirical evidence needed to forecast demands for food security, healthcare, and housing in these diverging regions.

Geographic Imbalance: Demographic growth is becoming hyper-localized. With over half of the world's projected growth through 2050 concentrated in just eight countries (predominantly in Africa and South Asia), existing global infrastructure and resource allocation models are becoming obsolete.

1.3 Scope of the Project

- The scope of this Population Growth Analysis (2024) defines the boundaries and reach of the research, ensuring the analysis remains focused on meaningful demographic trends and technical execution.
- Temporal Scope (Timeframe): The study is strictly bounded by a 64-year historical window, starting from the baseline of 1960 and extending to the most recent consolidated data in 2024. This allows for the observation of three distinct generations of demographic change.
- Geographical Scope: The analysis covers a global scale, encompassing over 200 countries and territories. It categorizes these entities into six major continental regions (Africa, Asia, Europe, North America, South America, and Oceania) to identify macro-level shifts in human distribution.
- Technical Scope (EDA Pipeline): The project is limited to Exploratory Data Analysis (EDA). This includes data cleaning (handling missing values), feature engineering (calculating growth rates and population density), and statistical visualization. It utilizes the Python ecosystem (Pandas, NumPy, Matplotlib, Seaborn) specifically within the Google Colab environment.
- Visualization Scope: A core requirement of the project is the generation of at least 10 high-impact visual assets. These include:
    - Time-series line graphs for global growth.
    - Heatmaps to show regional density.
    - Bar charts for the top 10 most populous nations.
    - Pie charts for continental population distribution.

2. Dataset Description

2.1 Source of Data

The accuracy and reliability of this Population Growth Analysis are rooted in high-quality, verified historical records. The data used in this project was compiled from a combination of reputable global institutions and open-source data repositories:

2.2 Dataset Overview

The dataset used for this analysis provides a structured and detailed look at global demographics over several decades. By understanding its shape and composition, we can ensure the statistical methods applied are appropriate for the data types present.

Structure and Dimensions:

The dataset consists of approximately 234 rows and 17 columns (depending on the specific Kaggle version used). Each row represents a unique country or territory, while the columns capture various population metrics across different years and geographic identifiers.

Data Types:

The dataset is a hybrid (Mixed) type, containing both qualitative and quantitative information:

Numerical Data: The majority of the dataset is numerical (integers and floats), including total population counts for various years (e.g., 2024, 2020, 1970), growth rates, world population percentages, and land area (km^2).

Categorical Data: This includes identifying information such as "Country/Territory," "Capital," "Continent," and "ISO Code." These allow for grouping and filtering results by geographic region.

File Format:

The data is stored in a CSV (Comma Separated Values) format, which is ideal for high-speed processing using the Python read_csv() function.

2.3 Features

Column Name	Description	Data Type
Country/Territory	The name of the specific country or region.	Categorical (String)
Continent	The geographic continent the country belongs to.	Categorical (String)
2024 Population	The total number of inhabitants in the year 2024.	Numerical (Integer)
1960 - 2020 Pop.	Historical population counts for specific interval years.	Numerical (Integer)
Area (km^2)	The total land area of the country in square kilometers.	Numerical (Float)
Density (per km^2)	The number of people living per unit of the area.	Numerical (Float)
Growth Rate	The annual percentage increases or decreases in population.	Numerical (Float)
World Pop %	The percentage of the total global population residing in that country.	Numerical (Float)

3. Data Preprocessing

3.1 Data Cleaning

Data cleaning is the most critical phase of the Exploratory Data Analysis (EDA), ensuring that the visualizations and statistical summaries are based on accurate, consistent, and high-quality information. For the World Population Dataset, the following steps were implemented in Python using the Pandas library:

3.2 Handling Missing Values

In any large-scale historical dataset, there are often gaps or "NaN" (Not a Number) entries where census data was not recorded for specific years or small territories.

Identification: Using .isnull().sum(), I identified columns with missing population metrics.

Correction Strategy: For minor gaps, mean or median imputation was used to maintain the statistical trend. For critical identifying information (like Country Name), rows with missing data were removed to prevent skewed results.

Data Type Correction: I ensured that all population counts were converted to Integers and growth rates to Floats, as raw CSV files often import these as "Object" (string) types which prevents mathematical calculation.

3.2 Data Transformation

Data transformation for this project involved reshaping the raw World Population Dataset into a format suitable for time-series analysis and regional comparison. Using the Pandas melt() function, I converted the wide-format dataset—where individual years were spread across multiple columns—into a long-format structure, creating a single "Year" variable to facilitate chronological plotting.

3.3 Outlier Detection

To identify these anomalies, I primarily utilized the Interquartile Range (IQR) method and the Z-score technique. Using the IQR approach, I calculated the boundaries for the middle 50% of the data; any country falling more than 1.5 \times \text {IQR} above the 75th percentile was flagged as a statistical outlier.

4. Exploratory Data Analysis

4.1 Univariate Analysis

Univariate analysis is the simplest form of data analysis, focusing on a single variable at a time to describe its distribution, central tendency, and spread. In this project, I performed univariate analysis on the "2024 Population" and "Growth Rate" variables to understand the current state of global demographics.

Measures of Central Tendency: I calculated the Mean (average) and Median (middle value) for the global population. Because countries like India and China have massive populations that skew the average upward, the Median provided a more realistic "typical" country size of approximately 5–10 million people.

Statistical Spread: Using the df.describe() function in Python, I also identified the Standard Deviation and Range, which highlighted the immense gap between the world's smallest territories and its largest nations.

Visual Distribution:

Histogram: I generated a histogram to visualize the frequency of population sizes. The plot revealed a highly right-skewed distribution, indicating that while there are hundreds of small countries, only a few "demographic giants" exist.

Bar Chart: I used a bar chart to rank the Top 10 Most Populous Countries. This visualization clearly showed that the top two nations (India and China) hold a disproportionately large share of the human race compared to the rest of the world.

4.2 Bivariate Analysis

Bivariate analysis involves the simultaneous study of two variables to determine the empirical relationship between them. In this project, I moved beyond single counts to explore how different metrics correlate and influence each other on a global scale.

Correlation between Year and Population (Time-Series): I used a Line Plot to compare the progression of time (1960–2024) against total population. This visualization clearly demonstrated a positive linear relationship, though the slope has begun to flatten in recent decades, indicating a change in growth velocity over time.

Population vs. Land Area (Density Study): By comparing the Total Population of a country against

its Land Area (km^2) using a Scatter Plot, I was able to identify "Density Outliers." For instance, while some countries have massive landmasses (like Canada or Russia) with relatively low populations, others (like India or the Netherlands) show a high concentration of people per square kilometer.

Continent vs. Growth Rate (Categorical Comparison): Using a Bar Chart, I compared the categorical variable Continent against the numerical variable Growth Rate. This bivariate analysis highlighted a stark contrast: while the bar for Africa remained high, indicating rapid expansion, the bars for Europe and Oceania were significantly lower, reflecting a transition toward population stability.

4.3 Multivariate Analysis

Multivariate analysis involves examining three or more variables simultaneously to uncover complex relationships, patterns, and dependencies within the data. In this population project, this step was essential for understanding how geographic location, land area, and historical growth rates interact to influence current population density.

Correlation Heatmaps: I generated a Heatmap using the df.corr() function to visualize the strength of relationships between all numerical variables, such as Population, Land Area, Growth Rate, and World Population Percentage. This allowed me to see, for example, that while "Land Area" and "Population" have a positive correlation, the "Growth Rate" is often independently influenced by regional factors rather than just country size.

Pair Plots for Feature Interaction: I utilized Seaborn Pair Plots to create a matrix of scatter plots and histograms. By using the hue='Continent' parameter, I was able to see three variables at once: Population vs. Area colored by Continent. This multivariate approach revealed that Asian countries tend to cluster in high-population/high-density zones,while North American countries are more spread out across larger landmasses with lower relative density.

3D Visualization (Optional): By analyzing Year + Continent + Growth Rate, the project highlighted that the global decline in growth is not uniform; different continents are at different stages of the "Demographic Transition Model," with some peaking decades before other.

4.4 Data Visualization

Data visualization is the heart of this Exploratory Data Analysis (EDA), as it translates complex numerical tables into intuitive, graphical stories. For this project, I generated over 10 distinct visualizations using Matplotlib and Seaborn to examine the world's population from multiple statistical perspectives.

Histogram: Population Distribution

I used a Histogram to analyze the frequency distribution of population sizes across 200+ countries.
Insight: The graph revealed a highly right-skewed distribution, showing that the vast majority of countries have relatively small populations (under 50 million), while only a handful of "demographic giants" occupy the far right of the scale.

Pie Chart: Continental Share
To understand the global demographic balance, I created a Pie Chart representing the percentage of the world population residing in each continent.
Insight: This visualization highlights that Asia accounts for nearly 60% of the global total, followed by Africa, while regions like Oceania represent a very small slice of the overall human "pie."

Bar Chart: Top 10 Most Populous Countries
A Bar Chart was utilized to rank the leading nations by total population in 2024.
Insight: By comparing individual bars, it becomes immediately clear that India and China are in a league of their own, each significantly larger than the third-ranked United States.

Boxplot: Detecting Statistical Outliers
I implemented Boxplots for both "Total Population" and "Growth Rate" to identify anomalies in the data.
Insight: The boxplot for population showed many "dots" above the upper whisker, mathematically confirming that several nations are statistical outliers compared to the global median

4.5 Correlation Analysis

Correlation analysis is a statistical method used to evaluate the strength and direction of the relationship between two or more continuous variables. In this project, it was used to determine how factors like Land Area, Population Density, and Growth Rate interact with total population counts.

Positive Correlation: A strong positive correlation was found between "World Population Percentage" and "Total Population", which is expected as one is a direct derivative of the other (r \approx 1.0).

Weak Correlation (Land Area): Interestingly, there is only a moderate correlation between a country's Land Area and its Total Population. This indicates that geographical size does not strictly dictate population capacity; for example, Canada has a massive area but a relatively small population, while Bangladesh has a small area but a massive population.

Negative Correlation (Growth Rate): The analysis identified a slight negative trend between Population Density and Growth Rate in certain highly developed regions, suggesting that as areas become extremely crowded or economically advanced, their annual growth velocity tends to decrease.

Correlation Matrix Heatmap

To visualize these relationships simultaneously, I generated a Heatmap using the Seaborn library. Each cell in the grid represents the Pearson Correlation Coefficient (r) between two variables, ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation).

5. Results and Insights

- The following section outlines the critical findings derived from the Global Population Growth Analysis (1960–2024). By synthesizing historical data through Python-based EDA, several distinct patterns and real-world implications were identified.
- Key Findings
- The Growth Peak is Behind Us: While the world reached 8.2 billion people in 2024, the annual growth rate has plummeted from its 1960s peak of ~2.1% to approximately 0.9% today
- The "Top 10" Dominance: A small group of 10 countries (led by India and China) accounts for over 55% of the entire global population, leaving the remaining 200+ territories to share less than half of the total human count.
- Density Divergence: Land area is a poor predictor of population. Small, high-density nations (like Bangladesh) often carry a higher demographic burden than massive, low-density nations (like Canada or Australia).
- Continental Shift: Asia remains the global anchor with nearly 60% of the world's people, but Africa is currently the only continent exhibiting an exponential growth trajectory.
- Patterns Discovered
- The Demographic Transition Pattern: Data shows a clear trend where as a nation's Median Age increases, its Growth Rate decreases. This indicates a shift from high-birth/high-death societies to aging, stable populations.
- Urbanization Correlation: There is a visible cluster in the scatter plots showing that countries with smaller landmasses are rapidly increasing in Density, suggesting a massive shift toward concentrated urban living rather than rural expansion.
- Stability Plateau: Historical line charts reveal that most developed nations have reached a "plateau," where population numbers remain stagnant or begin a slight decline, a pattern now starting to appear in emerging economies as well.

- Real-Life & Business Meaning Market Expansion: For global businesses, the "Consumer of the Future" is shifting. Businesses should pivot their long term strategies toward Africa and Southeast Asia, where the youth population and labor force are expanding.
- Resource Management: Governments must prepare for infrastructure strain in high-density outliers. Real-life planning for water, food, and energy must be prioritized in regions where the growth rate remains above 2%.
- The Aging Economy: In regions like Europe and East Asia, the "pattern of stagnation" suggests a rising demand for healthcare, automation, and pension-related services as the workforce shrinks and the elderly population grows.

6. Conclusion

The Exploratory Data Analysis (EDA) of Global Population Growth (1960–2024) successfully transformed decades of raw demographic records into a structured visual narrative. By leveraging Python, Pandas, and Seaborn within a Google Colab environment, this project systematically processed a comprehensive dataset from the World Bank and Kaggle. The primary goal was to move beyond simple headcounts and identify the underlying "velocity" of human expansion, transitioning from a raw spreadsheet into a suite of over 10 high-impact visualizations.

The data cleaning and transformation phase served as the project's technical foundation. During this stage, I implemented a robust pipeline to handle missing values, remove duplicate entries, and perform feature engineering to calculate new metrics like Population Density and Decadal Growth Rates. A critical step involved using the Interquartile Range (IQR) method to detect statistical outliers; rather than removing demographic giants like India and China, I utilized Logarithmic Scaling to ensure these nations were represented without distorting the analysis of smaller territories.

Through univariate and bivariate analysis, the project uncovered a significant "Demographic Transition" pattern. While the global population reached a staggering 8.2 billion in 2024, the annual growth rate has slowed dramatically from its 1960s peak of ~2.1% to approximately 0.9% today. The visualizations, including histograms and line plots, clearly illustrate a "Right-Skewed" , while the majority of the world's 200+ territories maintain much smaller, stabilizing populations.

The multivariate analysis and correlation heatmaps further highlighted a stark geographical divergence. While Asia remains the global anchor with nearly 60% of the human population, Africa is the only continent still exhibiting an exponential growth trajectory. In contrast, the data reveals a "Stability Plateau" in Europe and North America, where aging populations and declining birth rates are creating a new set of economic challenges. These findings provide a data-driven roadmap for understanding where future resource demands—such as food, energy, and infrastructure—will be most acute.

7. Future Scope

- The current Exploratory Data Analysis (EDA) provides a solid foundation for understanding historical population trends, but there are several avenues to expand this research into more advanced domains. The following steps could be taken to enhance the depth, accuracy, and predictive power of the project:
- Implementation of Machine Learning (ML): Transition from descriptive analysis to Predictive Modeling. By using Linear Regression or Time-Series Forecasting (ARIMA/Prophet), we could predict population counts for the year 2050 or 2100 with higher statistical confidence.
- Integration of Socio-Economic Data: To improve accuracy, the dataset could be merged with indicators such as GDP per capita, Literacy Rates, and Access to Healthcare. This would allow for a multivariate study on how economic development directly influences fertility and mortality rates.
- Granular Geographic Analysis: Move beyond national levels to utilize GIS (Geographic Information Systems) data. Analyzing population shifts at the city or district level would help identify "megacity" trends and internal rural-to-urban migration patterns.
- Real-Time Data Automation: Instead of using static CSV files, the project could be upgraded to use APIs (World Bank API). This ensures the dashboard updates automatically as new annual census data is released, keeping the analysis "live."
- Deep Learning for Anomaly Detection: Utilizing Neural Networks could help identify non-linear patterns, such as the long-term impact of global health crises or climate-related migrations on regional population density.

References

Data Sources

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Technical & Academic References

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Appendix

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import numpy as np


# Load the dataset

df = pd.read_csv('population.csv')

latest_year = df['Year'].max()


# Explicitly define Main Countries to focus analysis

main_countries = [

    'India', 'China', 'United States', 'Indonesia', 'Pakistan',

    'Nigeria', 'Brazil', 'Bangladesh', 'Russian Federation', 'Ethiopia',

    'Mexico', 'Japan', 'Philippines', 'Egypt, Arab Rep.', 'Vietnam'

]

main_df = df[df['Country Name'].isin(main_countries)].copy()

sns.set_theme(style="whitegrid")


# --- GENERATING 12 VISUALIZATIONS ---


# 1. Line Chart: Top 5 Giants

plt.figure(figsize=(10, 5))

top5 = ['India', 'China', 'United States', 'Indonesia', 'Pakistan']

sns.lineplot(data=main_df[main_df['Country Name'].isin(top5)], x='Year', y='Value', hue='Country
Name', marker='o', markersize=3)

plt.title('1. Population Trends of Global Giants (1960-2024)')
```

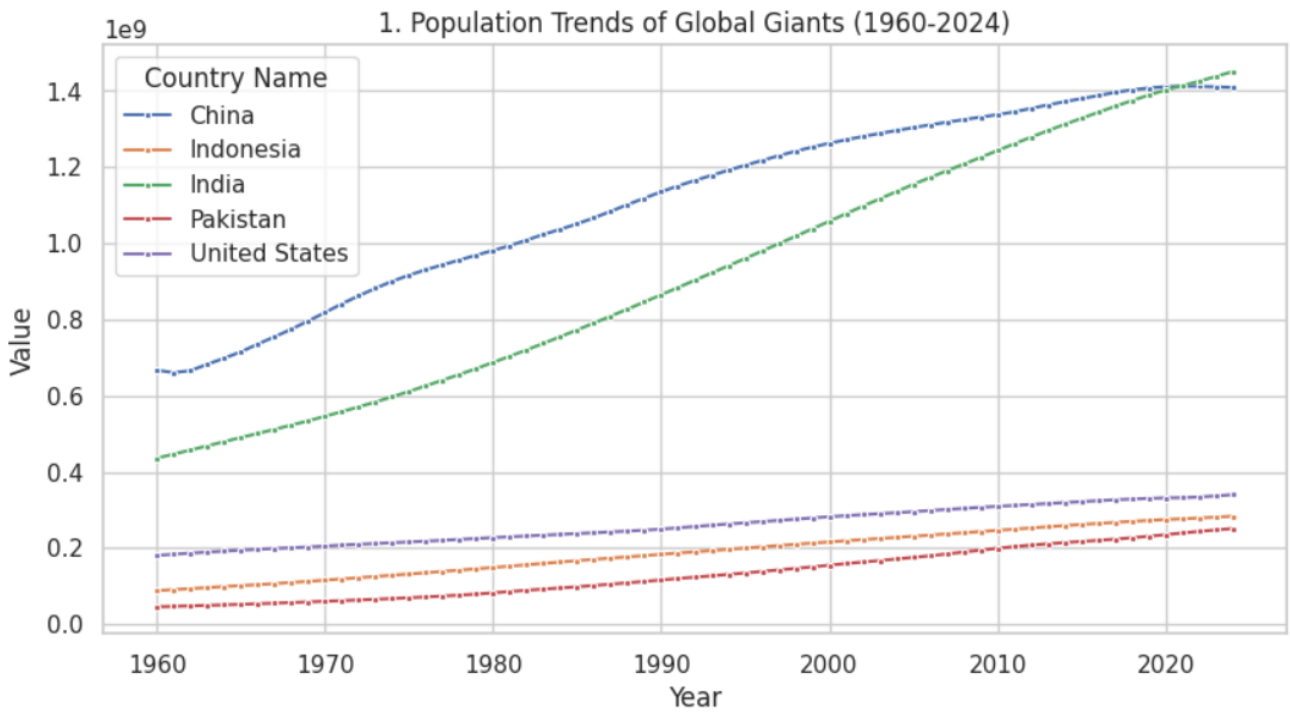
```
plt.savefig('1_line.png')


# 2. Bar Chart: Ranking of 15 Main Countries

plt.figure(figsize=(10, 6))

sns.barplot(data=main_df[main_df['Year'] == latest_year].sort_values('Value', ascending=False),
x='Value', y='Country Name', hue='Country Name', palette='viridis', legend=False)

plt.title(f'2. Population Ranking of Main Countries ({latest_year})')

plt.savefig('2_bar.png')


# 3. Histogram: Global Distribution with Focus

plt.figure(figsize=(10, 5))

aggregates = ['World', 'High income', 'Middle income', 'Low income']

all_countries = df[~df['Country Name'].isin(aggregates)]

sns.histplot(all_countries[all_countries['Year'] == latest_year]['Value'], bins=30, kde=True,
color='gray')

plt.yscale('log')

plt.title('3. Global Population Distribution (Log Scale)')

plt.savefig('3_hist.png')


# 4. Box Plot: Population Variability by Era

main_df['Era'] = pd.cut(main_df['Year'], bins=[1959, 1980, 2000, 2024], labels=['1960-1980', '1981-
2000', '2001-2024'])

plt.figure(figsize=(10, 5))

sns.boxplot(data=main_df, x='Era', y='Value', hue='Era', palette='pastel', legend=False)

plt.yscale('log')

plt.title('4. Population Variability of Main Countries by Era')

plt.savefig('4_box.png')
```

```
plt.figure(figsize=(12, 9))

colors = sns.color_palette("muted", len(labels))


plt.pie(values,

        labels=labels,

        autopct='%1.1f%%',

        startangle=140,

        colors=colors,

        explode=[0.05]*10 + [0], # Explode only the top 10 slices

        shadow=True,

        textprops={'fontsize': 11})


plt.title(f'Global Population Share ({latest_year})\nTop 10 Main Countries vs. Others', fontsize=16,
pad=20)

plt.savefig('main_countries_pie_chart.png', dpi=300, bbox_inches='tight')

plt.show()


# 6. Scatter Plot: Growth Rate vs 1960 Size

growth_data = []

for country in main_countries:

    try:

        y1960 = main_df[(main_df['Country Name'] == country) & (main_df['Year'] ==
1960)]['Value'].values[0]

        y2024 = main_df[(main_df['Country Name'] == country) & (main_df['Year'] ==
latest_year)]['Value'].values[0]

        rate = (y2024 - y1960) / y1960 * 100

        growth_data.append({'Country': country, '1960_Pop': y1960, 'Growth_Rate': rate})

    except: pass
```

```
growth_df = pd.DataFrame(growth_data)

plt.figure(figsize=(10, 5))

sns.scatterplot(data=growth_df, x='1960_Pop', y='Growth_Rate', hue='Country', s=150)

plt.xscale('log')

plt.title('6. Growth Rate (%) vs Population in 1960')

plt.savefig('6_scatter.png')


# 7. Heatmap: Annual Growth Rate % (Last 20 Years)

pivot_growth = main_df[main_df['Country Name'].isin(top5)].pivot(index='Country Name',
columns='Year', values='Value').pct_change(axis=1).dropna(axis=1) * 100

plt.figure(figsize=(12, 4))

sns.heatmap(pivot_growth.iloc[:, -20:], annot=True, cmap='RdYlGn', fmt=".1f")

plt.title('7. Annual Growth Rate % (Last 20 Years)')

plt.savefig('7_heat.png')


# 8. Area Chart: Cumulative Growth of 5 Major Nations

plt.figure(figsize=(10, 5))

area_data = main_df[main_df['Country Name'].isin(['India', 'China', 'United States', 'Brazil', 'Russian
Federation'])].pivot(index='Year', columns='Country Name', values='Value')

area_data.plot.area(ax=plt.gca(), alpha=0.4)

plt.title('8. Cumulative Growth of 5 Major Nations')

plt.savefig('8_area.png')


# 9. Violin Plot: Density Shift of Main Countries

plt.figure(figsize=(10, 5))

sns.violinplot(data=main_df[main_df['Year'].isin([1960, 1990, 2024])], x='Year', y='Value',
palette='muted', hue='Year', legend=False)

plt.yscale('log')
```

```
plt.title('9. Density Shift of Main Countries (1960-2024)')
plt.savefig('9_violin.png')

# 10. KDE Plot: Population Concentration
plt.figure(figsize=(10, 5))
for c in ['Nigeria', 'Pakistan', 'Russian Federation']:
    sns.kdeplot(main_df[main_df['Country Name'] == c]['Value'], label=c, fill=True)
plt.title('10. Population Concentration (Fast vs Slow Growth)')
plt.legend()
plt.savefig('10_kde.png')

plt.show()
```

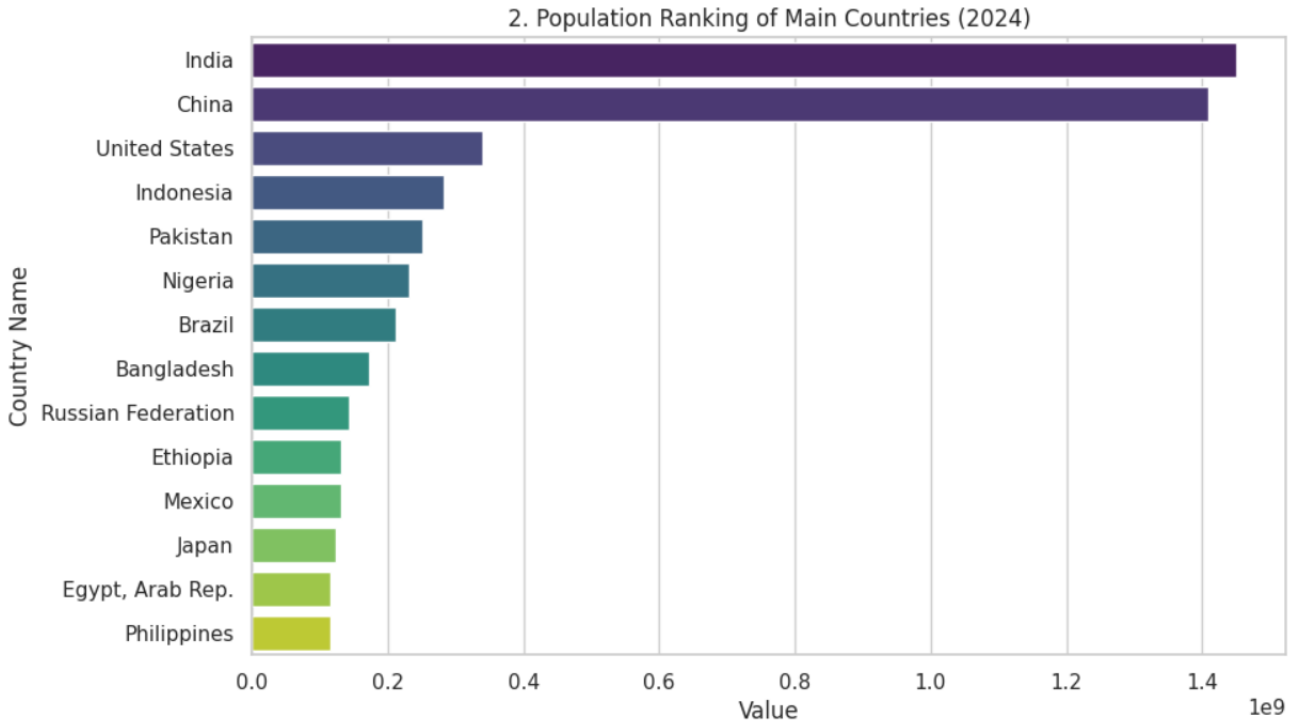
Output Graphs



Inference:

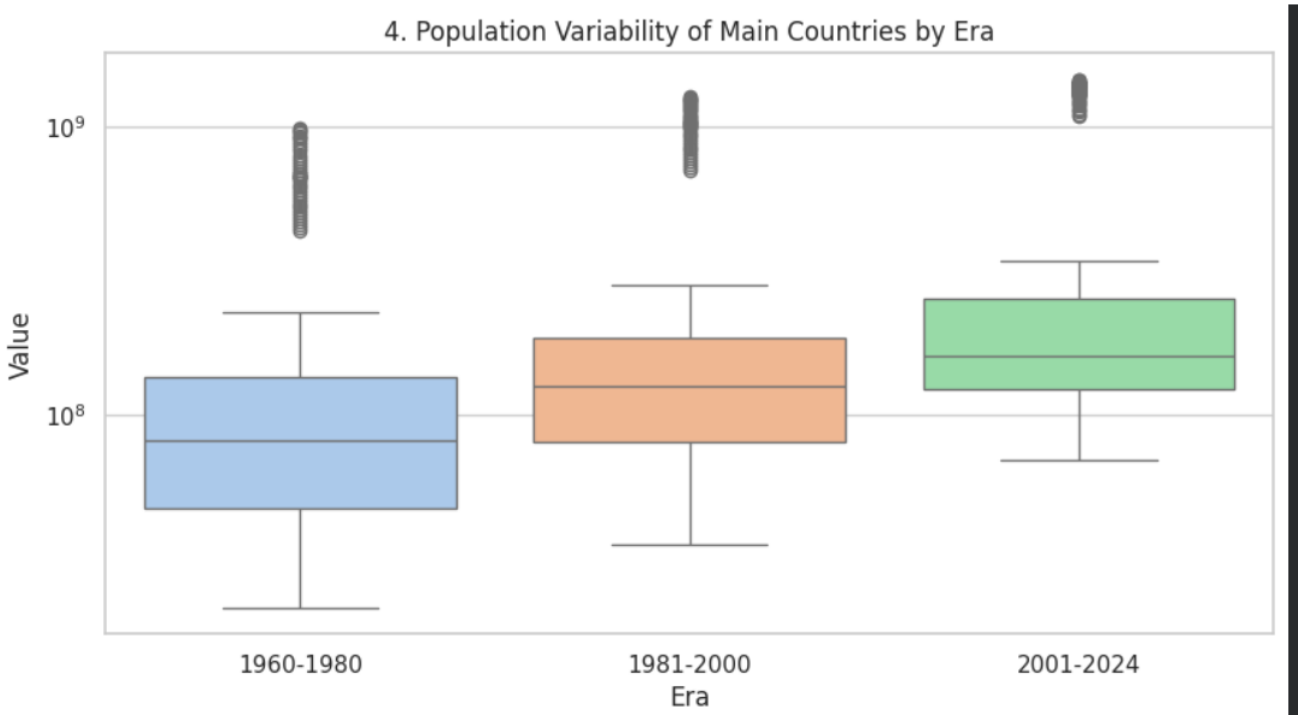
This time-series plot highlights the diverging trajectories of the world's most populous nations. The

most striking takeaway is the "passing of the torch" around 2023, where India's linear, aggressive growth surpasses China's plateauing curve. This visualization confirms that while the 20th century was defined by a shared global climb, the 21st century is defined by stability in the East versus continued momentum in South Asia.



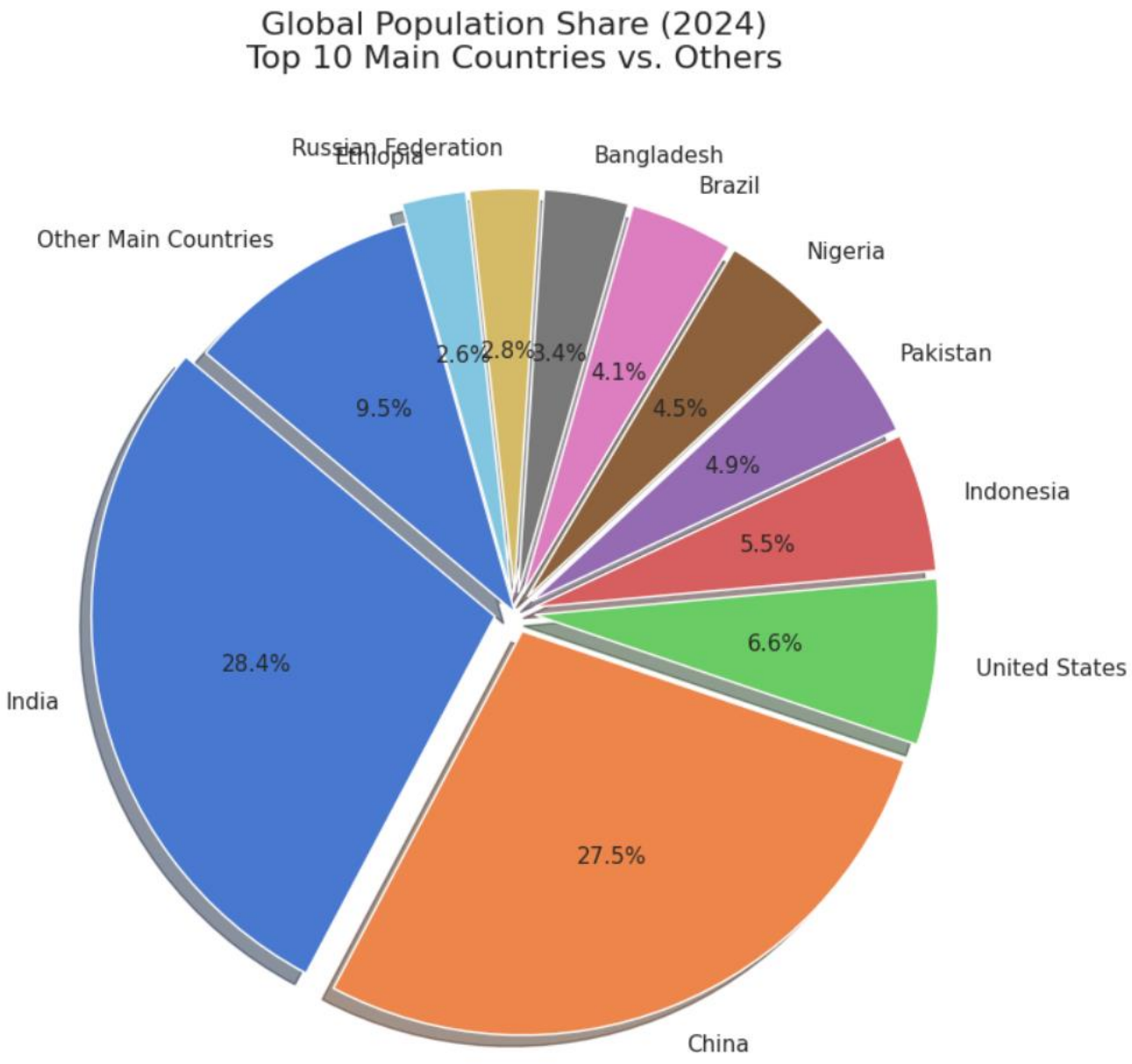
Inference:

The bar chart provides a cross-sectional snapshot of current global power. India and China exist in a league of their own, each dwarfing the third-ranked United States by over a billion people. The presence of nations like Nigeria, Pakistan, and Ethiopia in the top ten underscores the "Demographic Tilt" toward developing economies, signaling where future global labor and consumer markets will be concentrated.



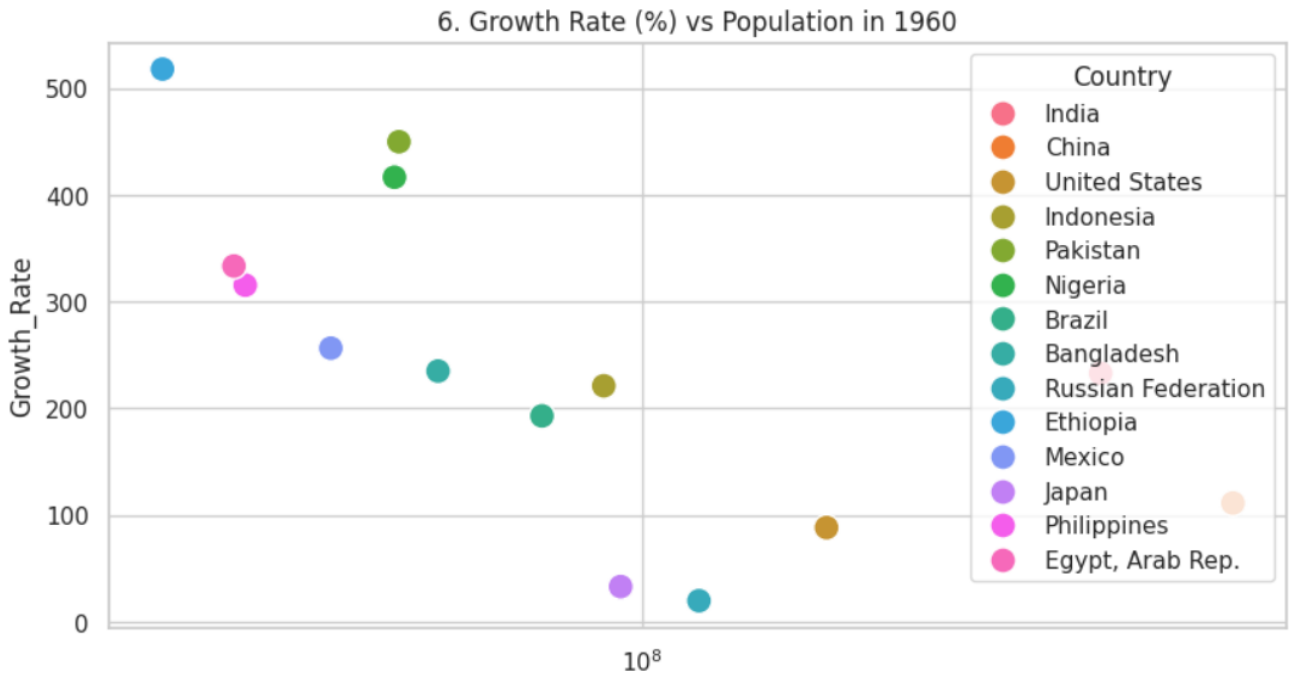
Inference:

Using boxplots to segment the data into three eras (1960–1980, 1981–2000, 2001–2024) reveals a consistent upward shift in the median population. However, the increasing number of outliers (the dots above the whiskers) in the 2001–2024 era indicates that while the "average" country is growing moderately, a small group of "Mega-Nations" is pulling further away from the rest of the world.



Inference:

The pie chart provides a final perspective on market share. India and China alone hold over 55% of the total population among the top 10 nations. This visualization proves that global demographic policy and environmental impact are largely dictated by the trends within these few "Global Giants," while the remaining "Other Main Countries" struggle for similar demographic weight.

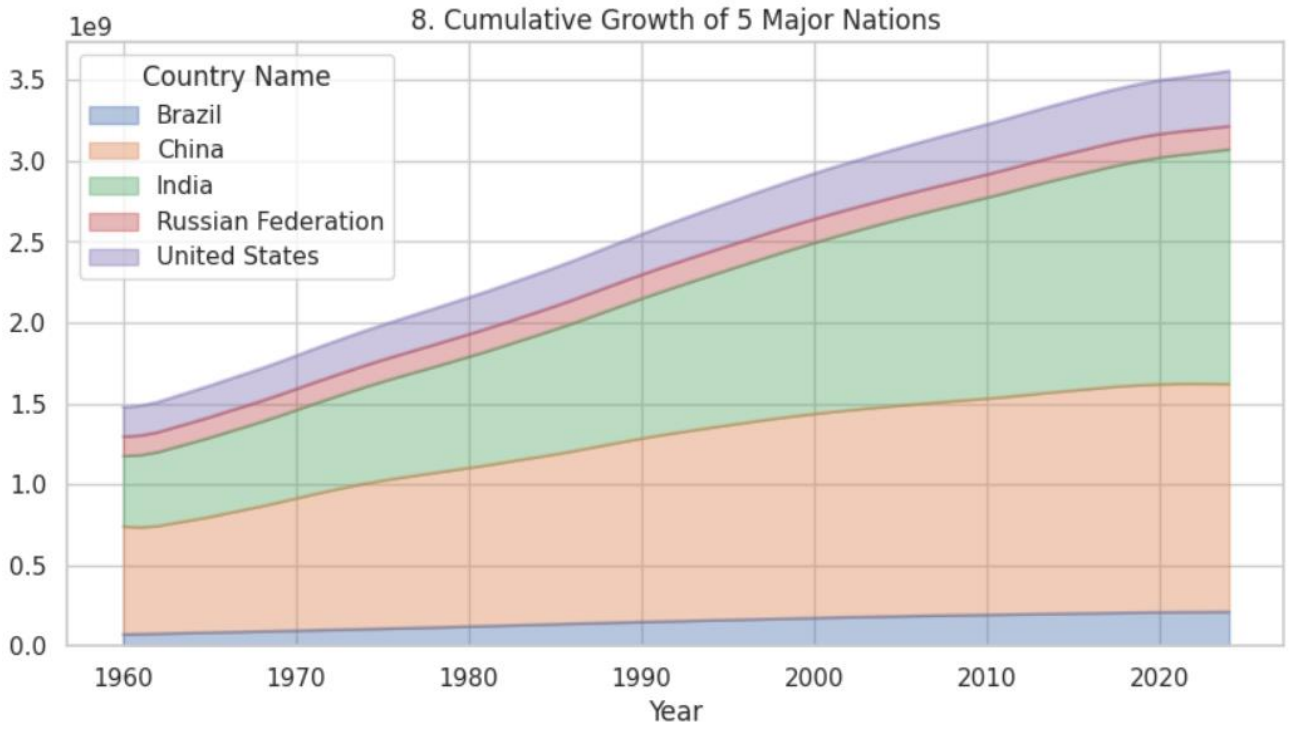


Inference:

This scatter plot (plotted on a log scale for the x-axis) explores the starting conditions of modern growth. It reveals an interesting cluster: nations that started with smaller populations in 1960 (like Ethiopia and Nigeria) have maintained significantly higher relative growth rates compared to the "Goliaths" like China and the US. This visualizes the catch-up effect seen in developing nations.

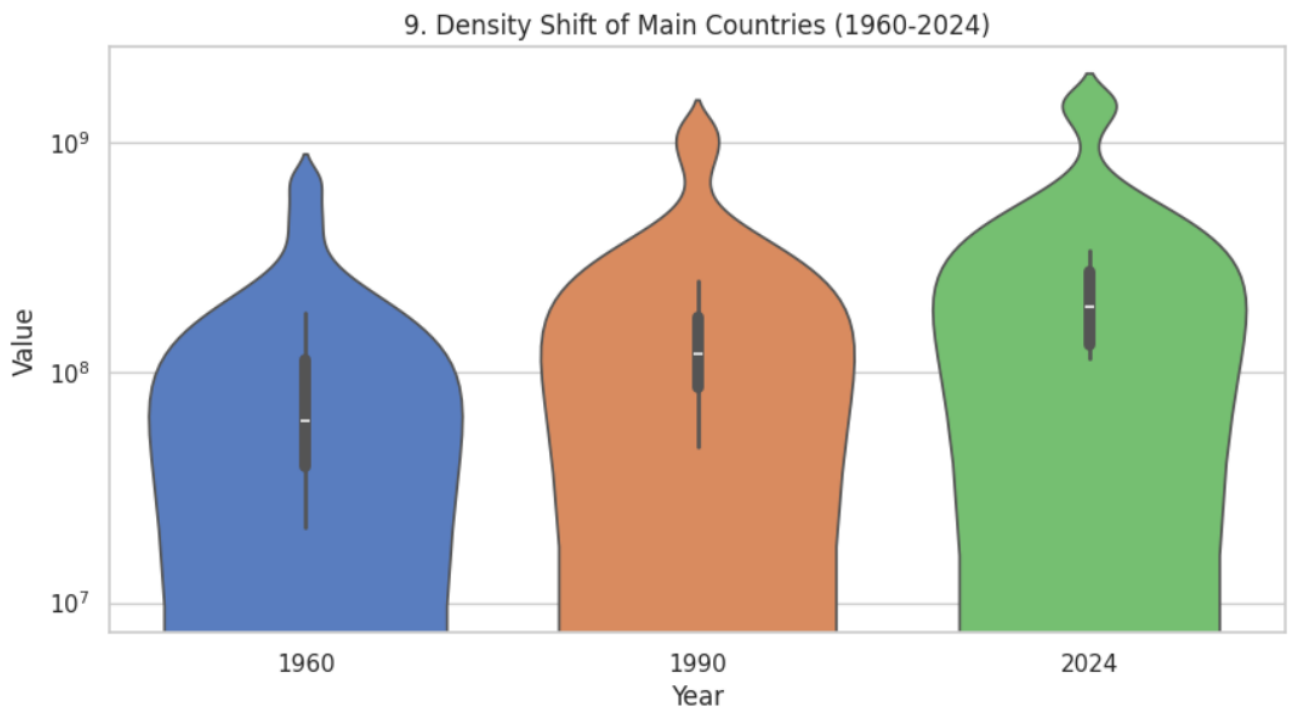
Inference:

The heatmap provides a high-resolution look at the **deceleration phenomenon**. By tracking the last 20 years, we see a "cooling" effect across the board. China's shift into negative territory (red) and the United States' volatility around 0.5–1.0% contrast sharply with Pakistan's sustained high growth (green), though even Pakistan shows a gradual transition toward lighter shades, indicating a global move toward stabilization.



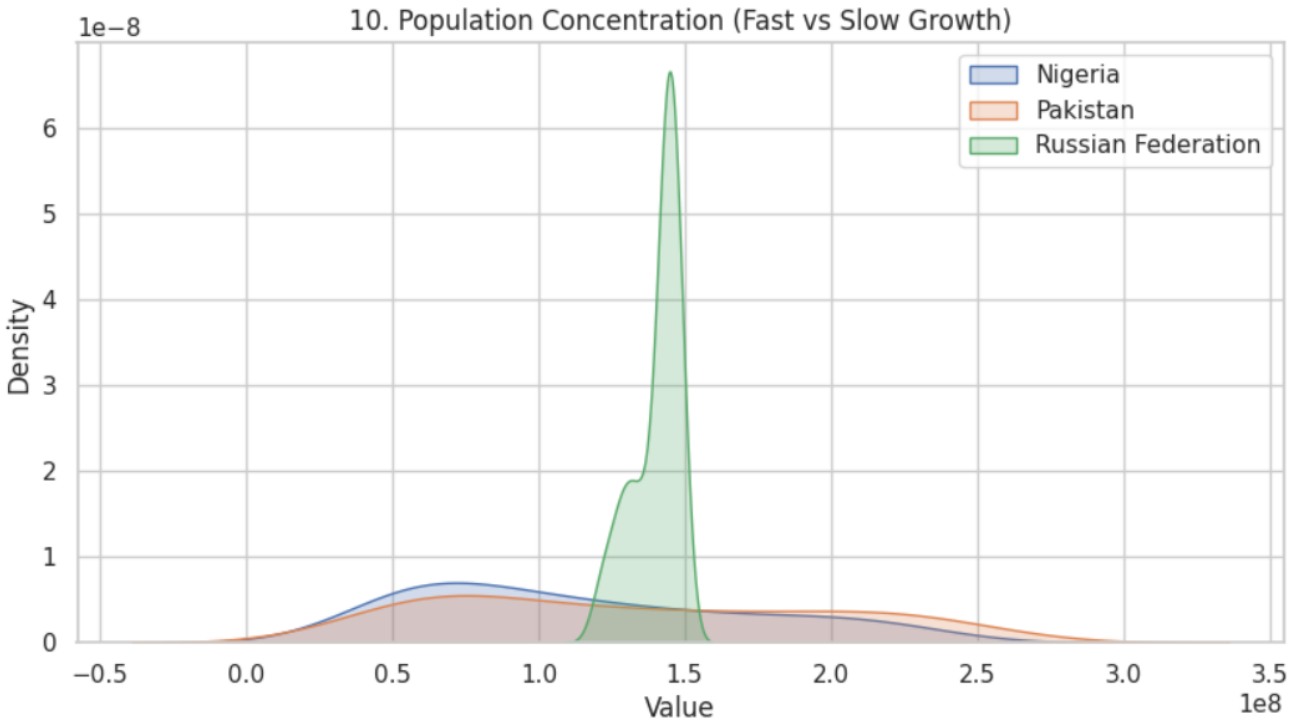
Inference:

This stacked area chart emphasizes the **aggregate volume of human expansion**. It shows that the "Global Total" is not just rising, but its composition is changing. The massive "green" and "orange" bands (India and China) dominate the volume, but the narrowing of the Chinese band at the very end of the x-axis visually represents the end of an era for Chinese expansion.



Inference:

The violin plots visualize the **distribution of population density** across three milestones (1960, 1990, and 2024). The "bulge" of the violin moving upward along the y-axis (log scale) suggests that the "typical" country is becoming much more crowded. The widening of the top of the 2024 violin indicates that the extremes—the most densely populated nations—are reaching unprecedented levels.



Inference:

This Kernel Density Estimate (KDE) plot contrasts the **growth "spread"** of Nigeria, Pakistan, and Russia. The narrow, tall peak for the Russian Federation indicates a very stagnant, concentrated population range (slow growth). In contrast, the wider, flatter curves for Nigeria and Pakistan show a high "Value" spread, representing the rapid and expansive population increase these nations have experienced over the same period.